

JetFighter Evaluation Protocol

1. Overview & Your Role

JetFighter is an automated ML pipeline that scans open-access scientific PDFs for figures that are inaccessible for people with color-vision deficiency (CVD). The pipeline has three stages:

- | | |
|------------------------------------|---|
| 1. Figure detection: | localises figures on each PDF page |
| 2. Colormap classification: | classifies each detected crop as <i>rainbow gradient</i> , <i>safe gradient</i> or <i>discrete plot</i> |
| 3. Accessibility check: | evaluates whether the color groups are distinguishable in grayscale (for discrete plots) |

Your job is to provide the **human ground truth** against which the automated pipeline is measured.

1.2. Validation Dataset

The validation dataset consists of ~150 recent open-access papers (published 2025-2026) sourced via OpenAlex. To ensure a diverse and balanced sample, around 10 papers were sampled across 15 different scientific fields, ensuring a broad representation of various journals. During the selection process, additional search criteria were applied to preferentially include papers containing multiple figures. At the same time, a small number of negative controls were intentionally included, consisting of papers that contained no figures at all.

1.3. Validation User Interface

You will use a streamlined User Interface (UI) to review image crops generated by the pipeline's figure detection module.

- **Left Panel:** Displays the original, full-color image crop.
- **Right Panel:** Displays the grayscale simulation of that exact crop.

All selections made in the UI are saved to a CSV file to be programmatically compared against the automated JetFighter predictions to calculate performance metrics.

2. Evaluation Protocol

For each presented crop, evaluate the image by progressing through the following steps:

2.1. Step 1: Detection Validation

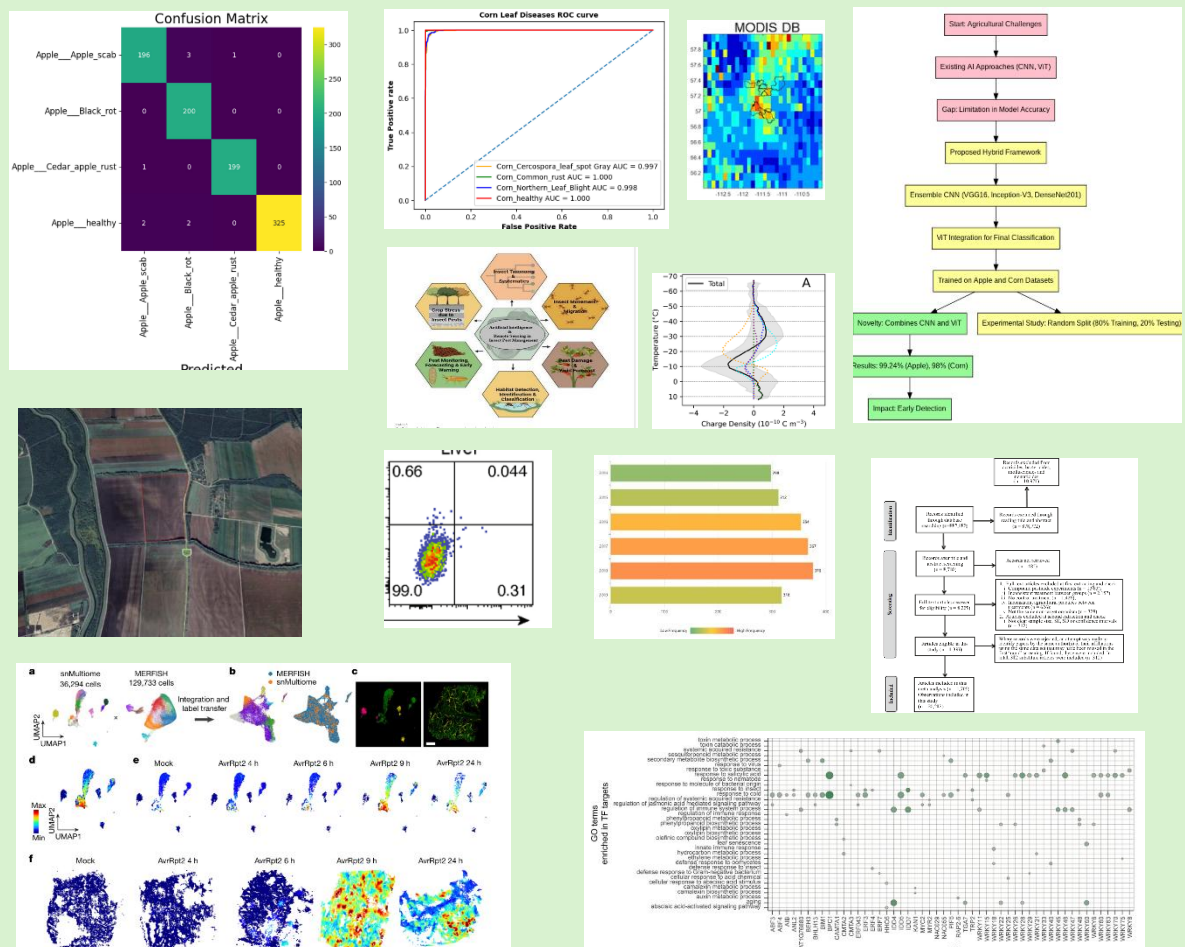
→ Assess whether the automated crop successfully captured a graphical data panel

Did the crop capture a valid scientific figure?

- The crop is a plot, chart, heatmap, diagram, (scientific) image, etc...
→ Yes
- The crop captures isolated text blocks, tables, publisher logos, standalone mathematical equations, blank space, etc...
→ No

Examples:

Valid scientific figures:



Invalid scientific figures:

$$\text{Accuracy} = \frac{t_p + t_n}{t_p + f_p + f_n + t_n} \quad (2)$$

$$\text{Precision} = \frac{t_p}{t_p + f_p} \quad (3)$$

$$\text{Recall} = \frac{t_p}{t_p + f_n} \quad (4)$$

$$\text{F1 score} = \frac{2t_p}{2t_p + f_p + f_n} \quad (5)$$

Actinobacteriota↓, Acidobacteria↑

Bacilli, Acidobacteria, Thermoleophilum, Clostridia and Acidimicrobia ↓; Gemmatimonadetes, Bacteroidia, Alphaproteobacteria and Gammaproteobacteria ↑



Logos

Insect	Crop	Research parameter	Reference
Major pests	Coffee	Detection and identification	Lee and Tardaguila (2023)
Diamond back moth	Cabbage	Abundance forecasting	Kar et al. (2022)
Major pests	Tomato	Prediction of pest outbreak	Holinger et al. (2020)
Thrips	Tomato	Management	Gonzalez et al. (2022)
Whitefly	Tomato	Life stages assessment and identification	Leite and Coradi (2022)
Fruit fly	-	Species detection and identification	Murmu et al. (2022)
Aphid	Soybean	Population dynamics	Murmu et al. (2022)
Insect pests	-	Pest classification	Xia et al. (2023)
Insect pests	-	Insect activity	Rydman et al. (2022)
Whitefly	Cotton	detect crop symptoms, environmental conditions, and physical characteristics	Joscano-Miranda et al. (2022)
Aphid	-	Pest forecasting	Bucurba et al. (2021)
British shoot and fruit borer, Epilachna beetle	Brinjal	Detection, identification, and classification	Sakumar et al. (2023)
Aphid	-	Monitoring and forecasting	Bata et al. (2023)
Spontaneous	Soybean	Detection and forecasting	Park et al. (2023)
Locust	-	Monitoring and management	Klein et al. (2021)
Stored grain pest	Wheat	Grain damage	Salvati (2020)

What happens, if...

... the crop is almost identical to the previous image?

The pipeline extracts all bounding boxes independently. Overlapping boxes can result in you seeing the exact same figure twice, just cropped slightly differently. Click the **"Duplicate of Previous"** button. This will automatically copy your entire evaluation from the previous image and proceed to the next one. Do *not* label a duplicate as "No" just because you already evaluated it.

... the figure is partially cut off or badly cropped?

The bounding box might miss axis labels, parts of the legend or cut off the edges of the plot. However, the core color-coded data is still visible enough to classify it. Select **Yes** and evaluate it normally. Do not select "No" just because it is an imperfect crop (a better, wider crop of the same figure might appear as the very next image). Only select **No** if the crop entirely misses the actual data.

2.2 Step 2: Colormap classification

→ Assess how color is utilized to convey data in the original color image (Left Panel).

What type of colormap does the figure use?

- The figure uses a **continuous, multi-hue colormap** that cycles through the colors **purple → blue → cyan → green → yellow → orange → red**. An important, though not mandatory (!), indicator is the presence of a **legend displaying a rainbow-spectrum colormap**. Crucially, the **brightest colors** (like yellow and cyan) are located in the **middle of the spectrum**. This means the colormap **lacks a monotonic luminance** (=brightness) gradient.

→ Rainbow Gradient

- The figure uses a **perceptually uniform, continuous colormap**. In these colormaps, the **brightest color** (usually yellow) is located at **one extreme end** of the spectrum, and the **colors transition monotonically to the darkest color** at the other end.

→ Safe Gradient

- The figure uses **distinct, categorical colors** to **separate different experimental groups**, conditions, or data categories (e.g bar charts, line graphs, scatter plots with discrete categories etc..). This category also serves as the default for all figures that were classified as a valid figure ("Yes") in Step 1 and do not feature any color gradients (neither Rainbow nor Safe). This includes for example purely black-and-white figures (not tables, they should have been excluded in Step 1), standard photographs, microscopy images, or pictures of authors.

→ Discrete Plot

Note:

A figure does not necessarily need to display a smooth, blended visual gradient. As soon as color is used to represent a **measure of magnitude or intensity** (e.g concentration, temperature...) rather than a distinct category, it is classified as a gradient.

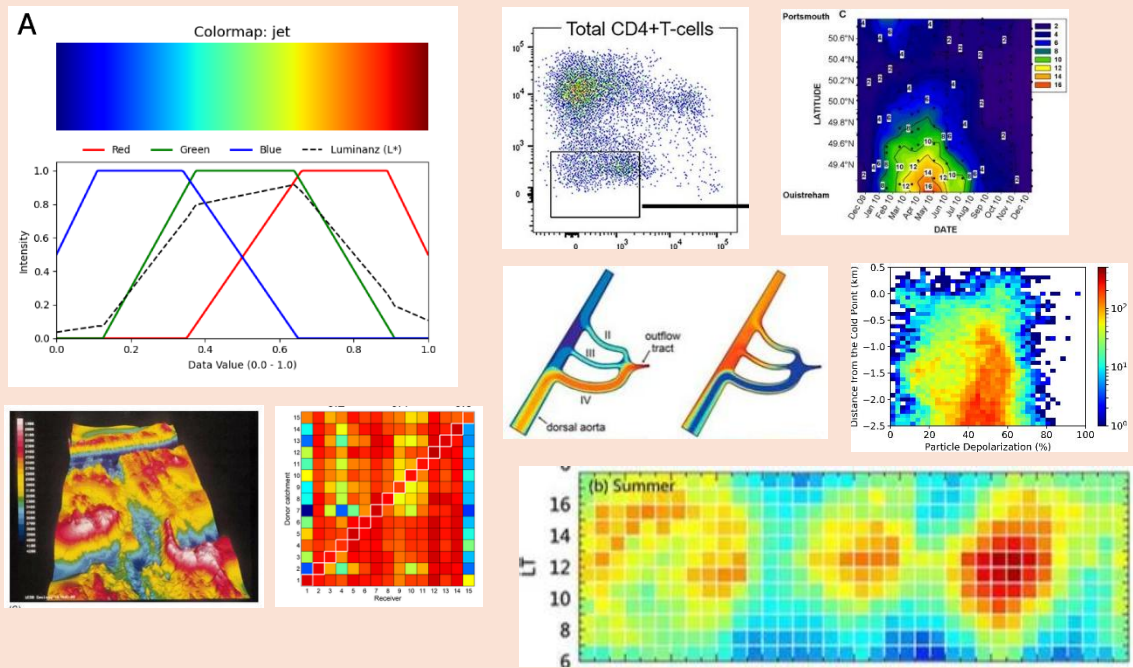
Useful link for colormap classification:

<https://matplotlib.org/stable/users/explain/colors/colormaps.html>

Examples:

Rainbow Gradient:

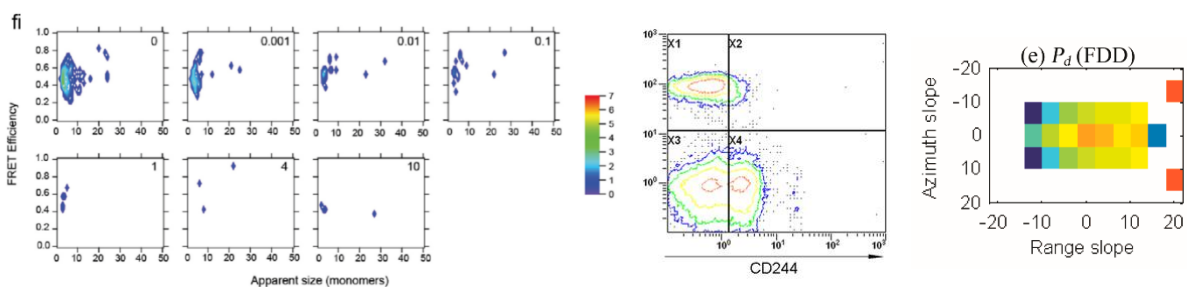
- purple → blue → cyan → green → yellow → orange → red
- **brightest colors** (like yellow + cyan) are located in the **middle of the spectrum**
- colormap **lacks a monotonic luminance (=brightness) gradient**



What happens, if...

... not all colors of the rainbow are visible in the figure?

The figure might only display a subset of the full rainbow spectrum, because the data values in that region do not span the entire range of the colormap. Choose **Rainbow Gradient**. The author likely used a full rainbow colormap (like "Jet") to create the figure.. Even if only a few hues are shown, they still belong to a problematic palette lacking monotonic luminance and will likely become indistinguishable in grayscale.



... the spectrum looks slightly different from the typical "Jet" colors?

The colormap cycles through multiple distinct hues, but the exact shades or transitions look slightly different from the classic "Jet" spectrum. Choose **Rainbow Gradient**. There are many multi-hue variants, but they all share the same fundamental flaw: they cycle through colors without a strict, continuous brightness gradient. This makes them universally problematic for colorblind accessibility and grayscale printing.

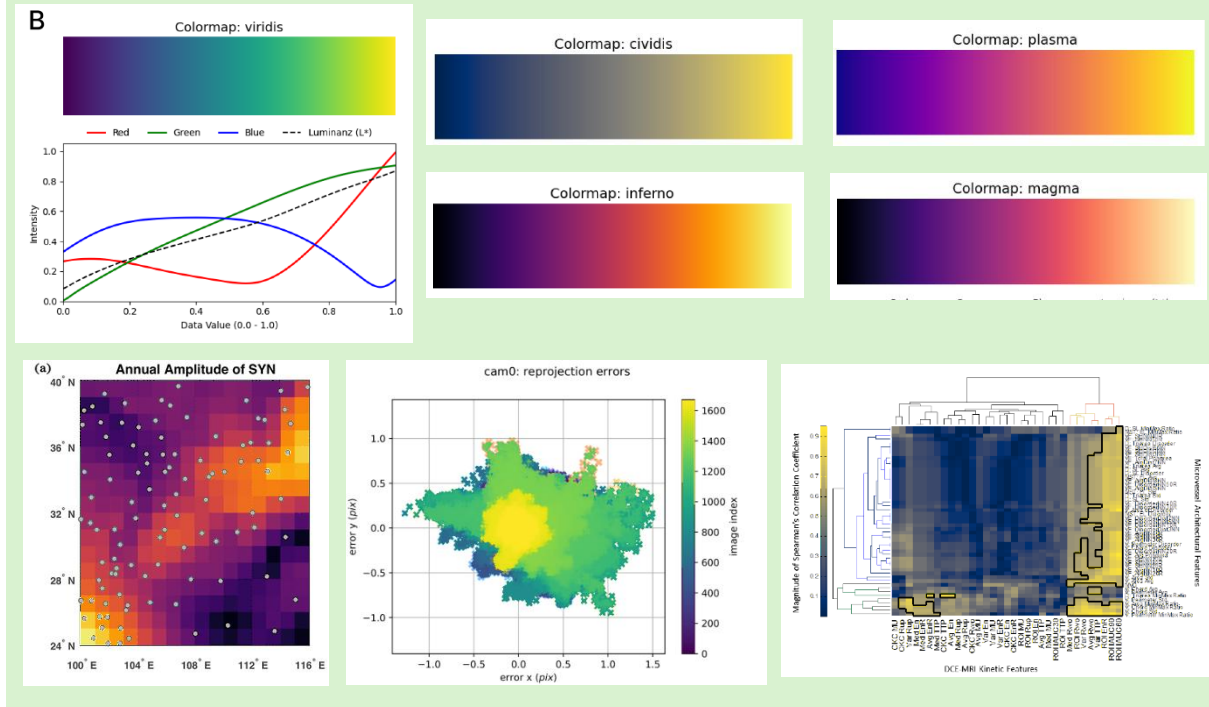


hsv



Safe Gradient:

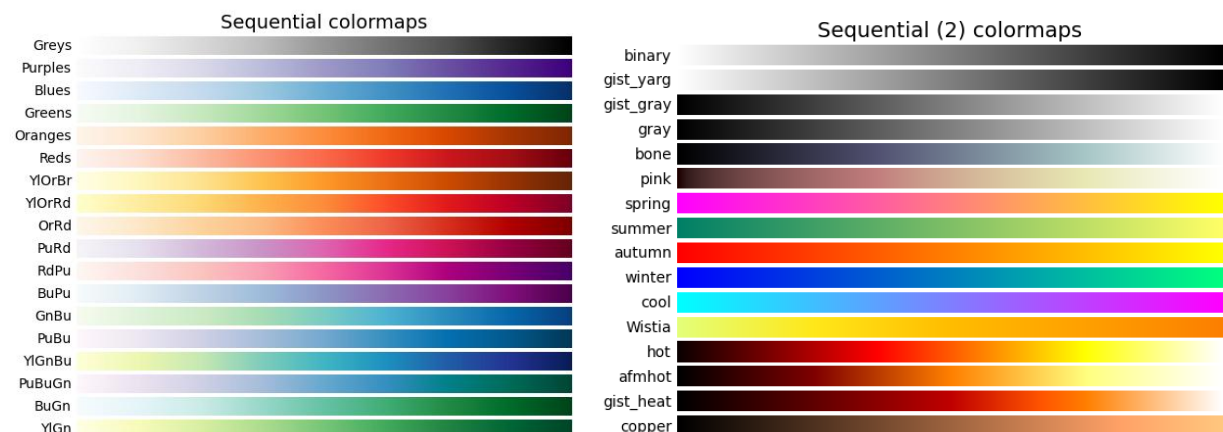
- **brightest color** (usually yellow) is located at **one extreme end** of the spectrum
- **colors transition monotonically to the darkest color** at the other end



What happens, if...

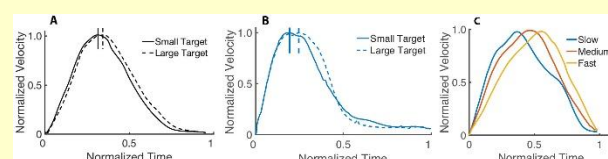
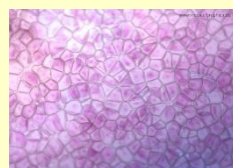
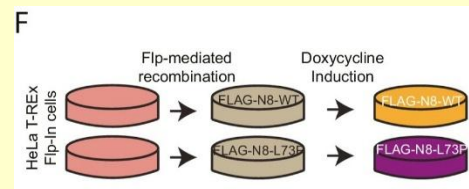
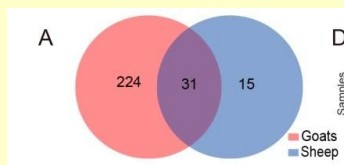
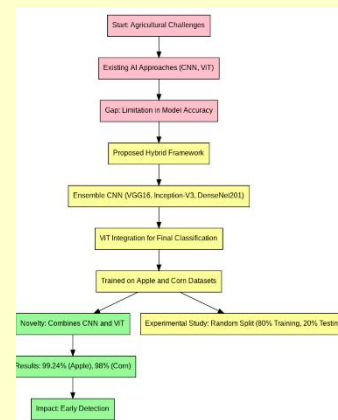
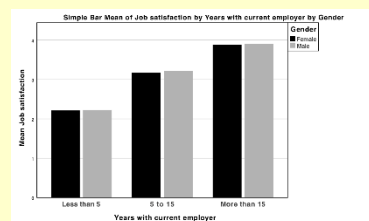
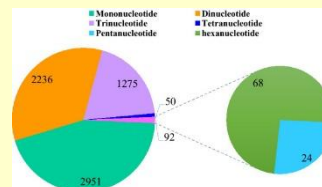
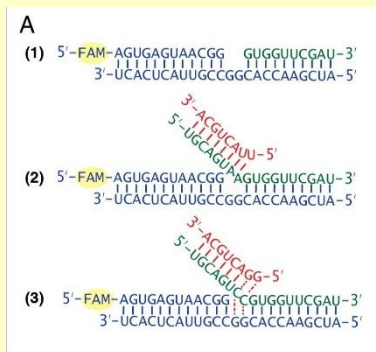
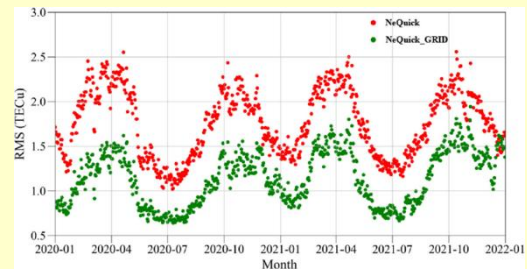
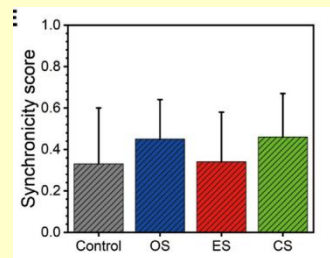
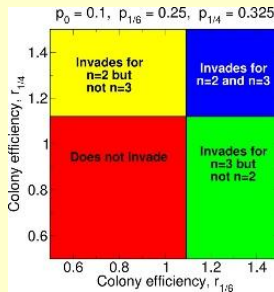
... the colormap uses only a single color fading from light to dark (or grayscale)?

The figure does not use multiple hues. Instead, it uses a single color that transitions from a light shade to a dark shade (e.g. light blue to dark blue) or a simple black-and-white grayscale gradient. Choose **Safe Gradient**. A single-hue or grayscale transition inherently relies on a strictly monotonic change in luminance, making it perfectly accessible for all readers.



Discrete Plot:

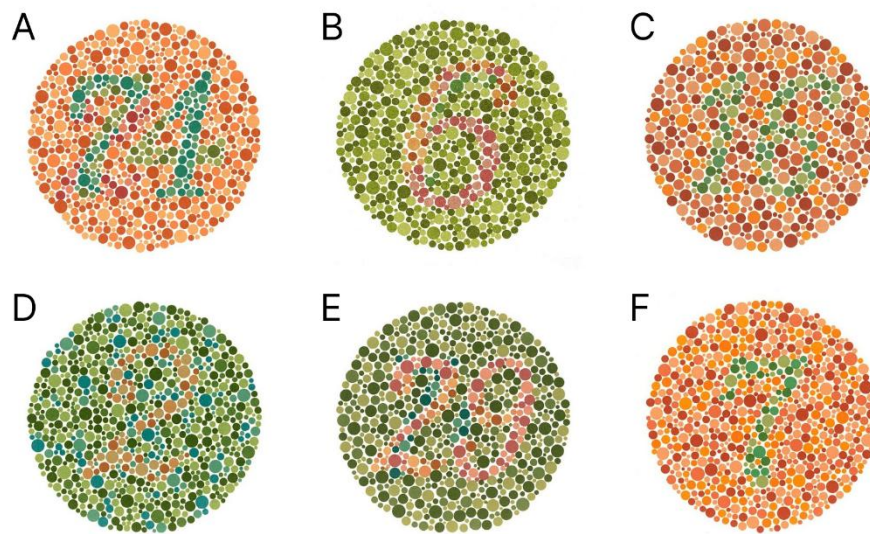
- figure uses **distinct, categorical colors** to **separate different experimental groups**
- bar charts, line graphs, scatter plots** (that do not use rainbow gradient!), **schemes, black-and-white figures, standard photographs, microscopy images, pictures of authors, etc...**



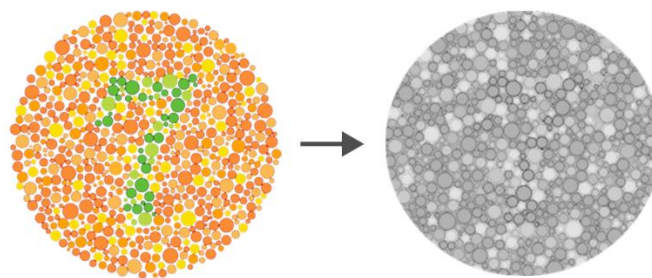
What happens if...

... the figure is an ishihara plate?

The image consists of a densely packed field of colored dots designed to hide a number or shape (a medical color vision test), or a similar pattern where color is used to separate distinct clusters. Choose **Discrete Plot**. Despite the chaotic and multi-colored appearance, the colors do not represent a continuous gradient. Instead, they encode strict categorical groups (e.g. the dots belonging to the number vs. the background dots).



Note: Ishihara plates are usually used in medical vision testing and are most of the time considered problematic in grayscale due to their reliance on color contrast, but still take a closer look and decide whether sufficient contrast remains in each case.



2.3 Step 3: Accessibility Check of Discrete Plots

→ Assess the accessibility of discrete plots by comparing the original image (left) with the grayscale simulation (right).

Are all data groups distinguishable in grayscale?

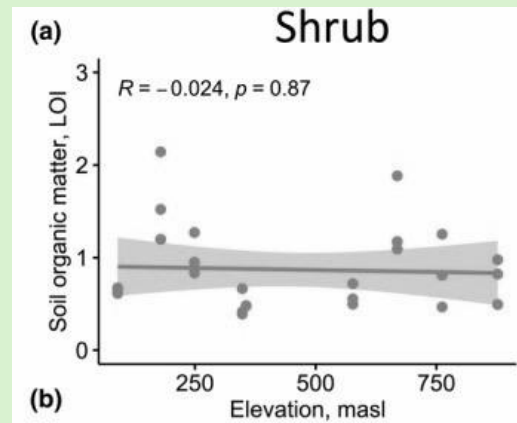
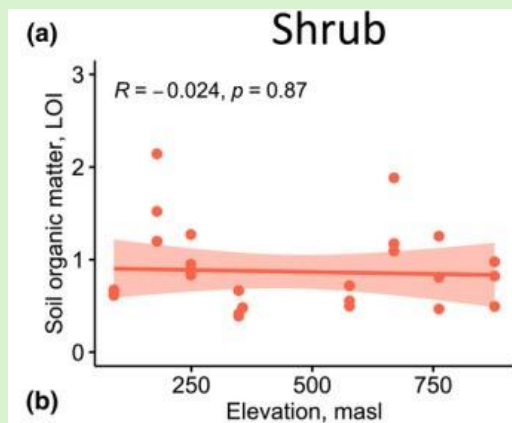
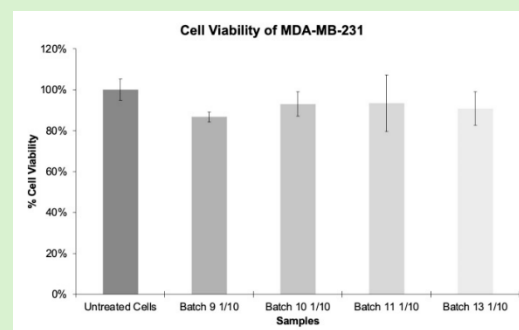
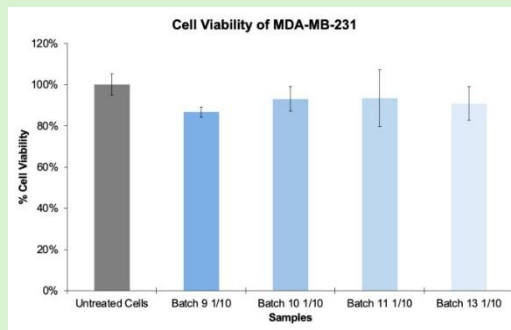
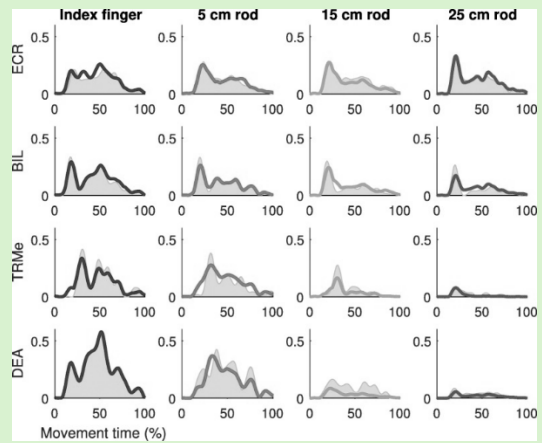
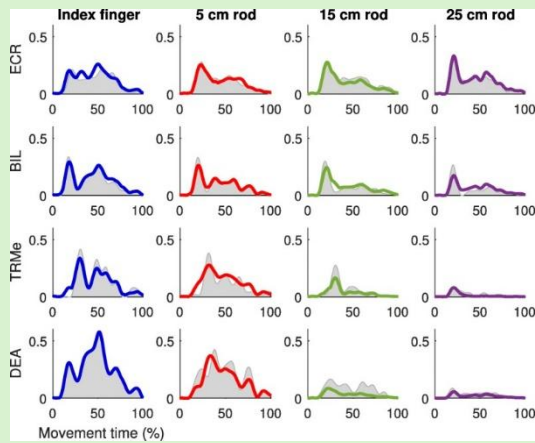
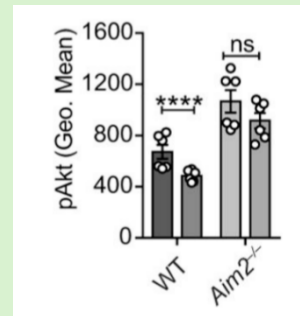
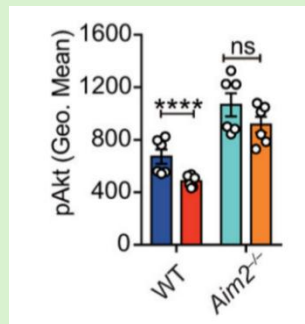
- In the grayscale simulation, every data group, line, or bar that needs to be compared remains clearly distinguishable by its gray level.
→ **Accessible**
- Two or more data categories merge into the same shade of gray in the simulation, making it difficult or impossible to distinguish them without color.
→ **Problematic**

Notes:

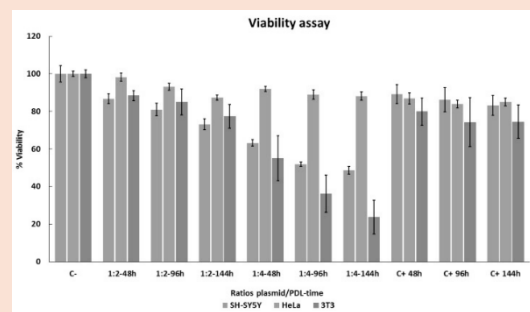
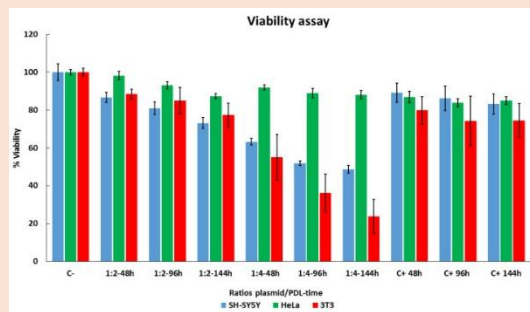
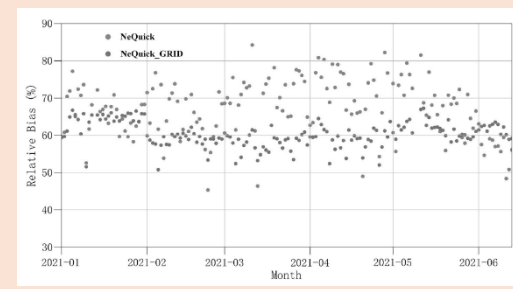
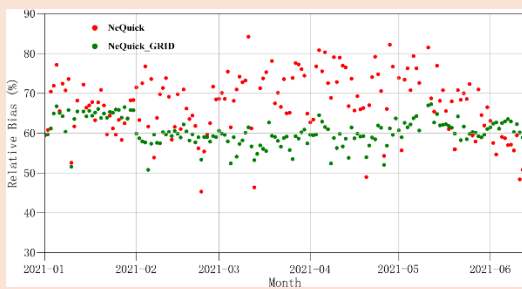
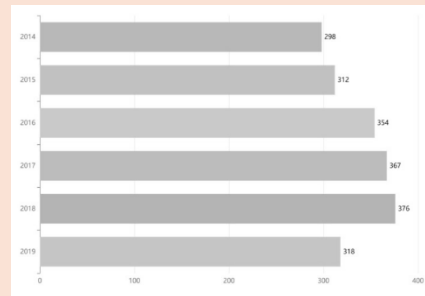
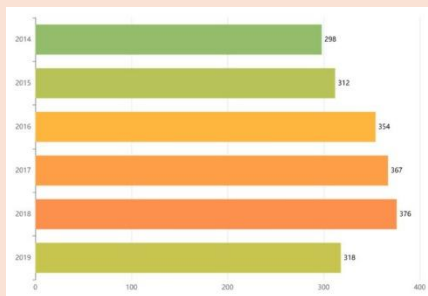
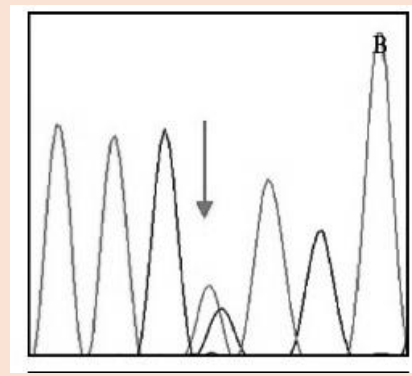
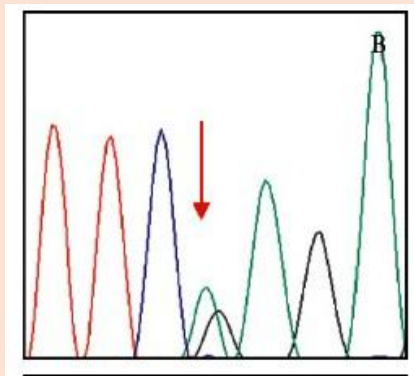
- Rely on your initial intuition. If the different data groups are clearly distinguishable at first glance, label it as **Accessible**. However, if you have to squint, strain your eyes, or carefully study the image to tell the gray levels apart (even if it is technically possible) label it as **Problematic**.
- Furthermore, judge the accessibility **purely on the difference in gray shades**, not on the fact that the data points might have different shapes (e.g triangles vs. circles) or are positioned far apart from each other. We are evaluating **color accessibility**.

Examples:

Accessible discrete plots:



Problematic discrete plots:



Edge Cases: Mixed Subfigures and Multi-Panel Crops

The JetFighter pipeline tries its best to detect and crop individual subfigures separately. However, sometimes a single crop might inevitably contain multiple subfigures that use entirely different colormaps or plot types. In these cases, the "**Worst-Case Rule**" applies: the more problematic category always overrules the accessible one.

What happens, if...

... a single crop contains multiple subfigures with different colormaps?

The crop mixes panel types (e.g., Panel A is a Jet heatmap, and Panel B is a simple bar chart or a Safe Gradient). Choose **Rainbow Gradient (in Step 2)**. The most severe, inaccessible colormap always dictates the label. If *any* of the subfigures in the crop uses a rainbow gradient, you must classify the entire image as a Rainbow Gradient, regardless of how much space the other plots take up.

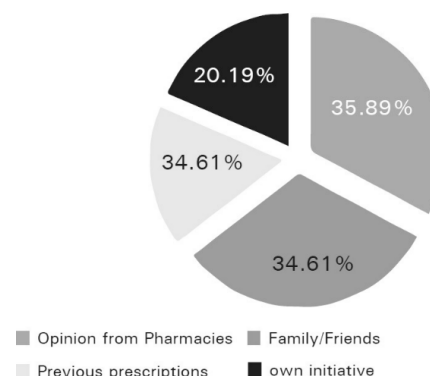
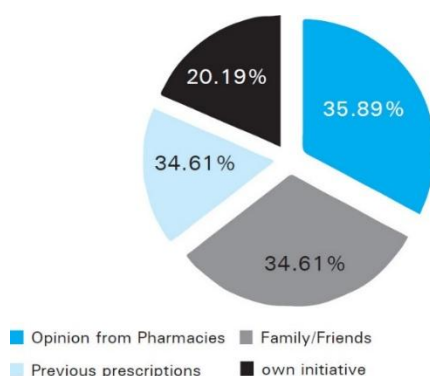
... a single crop contains multiple discrete plots, but only one is inaccessible?

The crop contains several distinct discrete plots (e.g three different bar charts). In the grayscale simulation, two of them are perfectly readable, but one of them loses its contrast and becomes ambiguous. Choose **Discrete Plot** and **Problematic (in Step 3)**. If the crop consists entirely of discrete plots, you must evaluate all of them in the grayscale simulation. If at least one subfigure is problematic in grayscale, you must label the entire crop as Problematic.

Edge Cases: Legends

... a plot remains distinguishable in grayscale, but the legend does not?

The plot itself contains bars, lines, or markers that remain clearly distinguishable in the grayscale simulation. However, the legend swatches become too similar in grayscale and can no longer be reliably matched to their corresponding data series. Choose **Problematic** (in Step 3). Accessibility requires that both the plot elements and the legend remain distinguishable. If the legend becomes ambiguous in grayscale, the entire figure must be classified as **Problematic**, even if the plot itself remains readable.



Decision Tree

